

### **Figure 1A: Hypoxic tumor region**

Photorealistic brightfield microscopy image of human breast cancer tissue, hematoxylin and eosin stain, 20x objective magnification. The tissue shows a dense tumor region with malignant epithelial cells arranged in irregular nests. Scattered throughout are macrophages with visibly elongated, spindle-shaped nuclei, particularly concentrated near hypoxic necrotic zones at the periphery of the tumor nests. Surrounding stromal regions show fibroblasts and collagen. Color palette typical of H&E: purple-blue nuclei, pink eosinophilic cytoplasm and extracellular matrix. No labels, no scale bar, no annotations. Scientific publication quality.

### **Figure 1B: Normoxic control region**

Photorealistic brightfield microscopy image of human breast cancer tissue, hematoxylin and eosin stain, 20x objective magnification. Dense tumor region with malignant epithelial cells in irregular nests. Scattered macrophages with round to oval nuclei, evenly distributed throughout the pink fibrous stroma. No necrotic zones, well-vascularized tissue with open capillary lumens visible. Color palette typical of H&E: purple-blue nuclei, pink eosinophilic cytoplasm and extracellular matrix. No labels, no scale bar, no annotations. Scientific publication quality.

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### **Figure 1C: High magnification inset showing elongated macrophage nuclei**

Photorealistic brightfield microscopy image of human breast cancer tissue, hematoxylin and eosin stain, 40x objective magnification, high magnification close-up. Focus on a cluster of tumor-associated macrophages with clearly elongated, spindle-shaped nuclei, visibly distinct from surrounding round tumor cell nuclei. Macrophages should have pale eosinophilic cytoplasm and elongated dark purple nuclei. Surrounding cells show round pleomorphic nuclei with prominent nucleoli. Rich pink stromal background with collagen fibers. No labels, no scale bar, no annotations. Scientific publication quality.

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### **Figure 1D: High magnification inset showing round macrophage nuclei (control)**

Photorealistic brightfield microscopy image of human breast cancer tissue, hematoxylin and eosin stain, 40x objective magnification, high magnification close-up. Focus on a cluster of tumor-associated macrophages with round to oval nuclei, similar in shape to surrounding lymphocytes but larger with more cytoplasm. Macrophages evenly interspersed among tumor cells. Surrounding malignant cells show round nuclei with

visible nucleoli. Pink collagen-rich stroma in background. No labels, no scale bar, no annotations. Scientific publication quality.

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#### **Figure 2A: IHC for macrophage marker (CD68) in hypoxic region**

This adds a second staining modality, which real papers always include to confirm cell identity:

Photorealistic brightfield microscopy image of human breast cancer tissue, immunohistochemistry staining for CD68 macrophage marker, 20x objective magnification. Brown DAB chromogen staining marking macrophage cytoplasm, hematoxylin counterstain showing blue-purple nuclei. Elongated spindle-shaped CD68-positive macrophages concentrated near pale necrotic tumor zones. Background tumor cells and stroma are unstained. No labels, no scale bar, no annotations. Scientific publication quality.

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#### **Figure 2B: IHC for hypoxia marker (HIF-1 $\alpha$ )**

Photorealistic brightfield microscopy image of human breast cancer tissue, immunohistochemistry staining for HIF-1 alpha hypoxia marker, 20x objective magnification. Brown DAB chromogen showing nuclear HIF-1 alpha positivity in tumor cells near necrotic zones. Strong brown nuclear staining in cells at the hypoxic periphery, weak or absent staining in well-oxygenated central tumor regions. Hematoxylin counterstain. No labels, no scale bar, no annotations. Scientific publication quality.